Beam Mismatch and Halo Formation

USPAS - Beam Physics with Intense Space Charge

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Derivation of equilibrium beam radius

The KV envelope equations describe the evolution of the elliptical beam radii $r_{x,y}$ in a linear focusing system:

$$
\begin{aligned}
r_x'' + \kappa_x(s) r_x - \frac{2Q}{r_x + r_y} - \frac{\varepsilon_x^2}{r_x^3} = 0, \\
r_y'' + \kappa_y(s) r_y - \frac{2Q}{r_x + r_y} - \frac{\varepsilon_y^2}{r_y^3} = 0,
\end{aligned}
$$

where $\varepsilon_{x,y}$ are the invariant emittances, Q is the beam perveance, and $\kappa_{x,y}(s)$ are the linear focusing strengths. We will study continuous, axisymmetric focusing systems with $\kappa_x(s) = \kappa_y(s) = k_0^2$. We will also assume equal emittances in each plane: $\varepsilon_x = \varepsilon_y = 0$. With these assumptions, the envelope equations become:

$$
\begin{aligned}
r_x'' + k_0^2 r_x - \frac{2Q}{r_x + r_y} - \frac{\varepsilon^2}{r_x^3} = 0, \\
r_y'' + k_0^2 r_y - \frac{2Q}{r_x + r_y} - \frac{\varepsilon^2}{r_y^3} = 0.
\end{aligned}
\tag{1}
$$

3D envelope modes in continuous focusing channels

It is natural to extend the above analysis to three-dimensional systems. We will simplify things by restricting our attention to upright beams with axial symmetry. We study the coupling between the transverse radius $r_\perp$ and the longitudinal radius r_z.

[...]

Particle-core models of halo formation

Introduction

Beam distributions in high-intensity accelerators tend to develop a low-density cloud of charge surrounding a dense, oscillating core, which we call a *halo*. For example, Figure 1 shows the measured two-dimensional phase space density after acceleration in the Spallation Neutron Source (SNS) linac. Although the halo density can be orders of magnitude less than the core, it still presents a problem in the SNS and other high-power facilities. The accelerator operating power is ultimately constrained by uncontrolled *beam loss*, which leads to radioactivation in the accelerator tunnel. High-power facilities must adhere to strict beam loss limits—typically set at 1 W/m. For a 1-MW beam, the loss limit corresponds to one out of every million particles, or a fractional loss of 10^{-6}. Once the loss limit is reached, increasing the beam power requires a proportional decrease in the beam loss. Understanding the mechanisms of halo formation is, therefore, highly relevant for modern and future high-power accelerators.